

Course Code and Title: MAT110 Quantitative Methods

Lesson 3: Week 4

Topic: Central Location and Dispersion

Introduction

Suppose you are given a data series. Someone asks you to tell some interesting facts about this data series. How can you do so? You can say you can find the mean, the median or the mode of this data series and talk about its distribution. But is it the only thing you can do? Are the central tendencies the only way by which we can get to know about the concentration of the observation? In this section, we will learn about another measure to know more about the data. Here, we are going to know about the measure of dispersion. As the name suggests, the measure of dispersion shows the scatterings of the data. It tells the variation of the data from one another and gives a clear idea about the distribution of the data. The measure of dispersion shows the homogeneity or the heterogeneity of the distribution of the observations.

Measures of Dispersion

Characteristics of Measures of Dispersion

- A measure of dispersion should be rigidly defined
- It must be easy to calculate and understand
- Not affected much by the fluctuations of observations
- Based on all observations

Classification of Measures of Dispersion

The measure of dispersion is categorized as:

1. An absolute measure of dispersion: The measures which express the scattering of observation in terms of distances i.e., range, quartile deviation. The measure which expresses the variations in terms of the average of deviations of observations like mean deviation and standard deviation.
2. A relative measure of dispersion: We use a relative measure of dispersion for comparing distributions of two or more data sets and for unit free comparison. They are the coefficient of range, the coefficient of mean deviation, the coefficient of quartile deviation, the coefficient of variation, and the coefficient of standard deviation.

Range

A range is the most common and easily understandable measure of dispersion. It is the difference between two extreme observations of the data set. If $x_{\max}$ and $x_{\min}$ are the two extreme observations then

$$\text{Range} = x_{\max} - x_{\min}$$

where $x_{\max}$ is the largest measurement in the data set and $x_{\min}$ is the smallest

Merits of Range

- It is the simplest of the measure of dispersion
- Easy to calculate
- Easy to understand
- Independent of change of origin

Demerits of Range

- It is based on two extreme observations. Hence, get affected by fluctuations
- A range is not a reliable measure of dispersion
- Dependent on change of scale

Example: Find the range of each data set.

Data Set I:	40	38	42	40	39	39	43	40	39	40
Data Set II:	46	37	40	33	42	36	40	47	34	45

Solution:

For Data Set I the maximum is 43 and the minimum is 38, so the range is $R = 43 - 38 = 5$.

For Data Set II the maximum is 47 and the minimum is 33, so the range is $R = 47 - 33 = 14$.

The range is a measure of variability because it indicates the size of the interval over which the data points are distributed. A smaller range indicates less variability (less dispersion) among the data, whereas a larger range indicates the opposite.

The Variance and the Standard Deviation

The other two measures of variability that we will consider are more elaborate and also depend on whether the data set is just a sample drawn from a much larger population or is the whole population itself (that is, a census).

Definition

The **sample variance** of a set of n sample data is the number s^2 defined by the formula

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

which by algebra is equivalent to the formula

$$s^2 = \frac{\sum x^2 - \frac{1}{n}(\sum x)^2}{n-1}$$

The **sample standard deviation** of a set of n sample data is the square root of the sample variance, hence is the number s given by the formulas

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}} = \sqrt{\frac{\sum x^2 - \frac{1}{n}(\sum x)^2}{n-1}}$$

Although the first formula in each case looks less complicated than the second, the latter is easier to use in hand computations, and is called a shortcut formula.

Example: Find the sample variance and the sample standard deviation of Data Set II.

Data Set I:	40	38	42	40	39	39	43	40	39	40
Data Set II:	46	37	40	33	42	36	40	47	34	45

Solution:

To use the defining formula (the first formula) in the definition we first compute for each observation x its deviation from the sample mean. Since the mean of the data is , we obtain the ten numbers displayed in the second line of the supplied table.

x	46	37	40	33	42	36	40	47	34	45
$x - \bar{x}$	6	-3	0	-7	2	-4	0	7	-6	5

$$\Sigma(x - \bar{x})^2 = 6^2 + (-3)^2 + 0^2 + (-7)^2 + 2^2 + (-4)^2 + 0^2 + 7^2 + (-6)^2 + 5^2 = 224$$

$$s^2 = \frac{\Sigma(x - \bar{x})^2}{n-1} = \frac{224}{9} = 24.\bar{8}$$

$$s = \sqrt{24.\bar{8}} \approx 4.99$$

The student is encouraged to compute the ten deviations for Data Set I and verify that their squares add up to 20, so that the sample variance and standard deviation of Data Set I are the much smaller numbers $s^2 = 20/9 = 2.22$ and $s =$

If the data set comprises the whole population, then the population standard deviation, denoted σ (the lowercase Greek letter sigma), and its square, the population variance σ^2 , are defined as follows.

Definition

The **population variance** and **population standard deviation** of a set of N population data are the numbers σ^2 and σ defined by the formulas

$$\sigma^2 = \frac{\Sigma(x - \mu)^2}{N} \quad \text{and} \quad \sigma = \sqrt{\frac{\Sigma(x - \mu)^2}{N}}$$

Note that the denominator in the fraction is the full number of observations, not that number reduced

by one, as is the case with the sample standard deviation. Since most data sets are samples, we will always work with the sample standard deviation and variance.

Finally, in many real-life situations the most important statistical issues have to do with comparing the means and standard deviations of two data sets. Figure 2.0 "Difference between Two Data Sets" illustrates how a difference in one or both of the sample mean and the sample standard deviation are reflected in the appearance of the data set as shown by the curves derived from the relative frequency histograms built using the data.

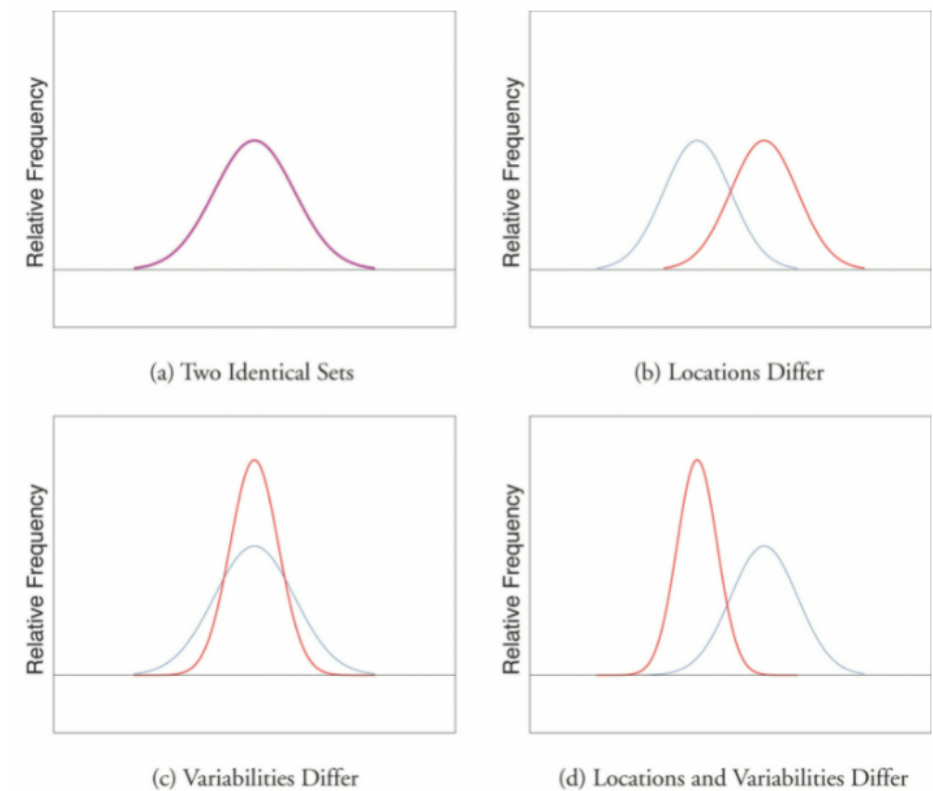


Figure 2.1 Difference between Two Data Sets

Relative Position of Data

When you take an exam, what is often as important as your actual score on the exam is the way your score compares to other students' performance. If you made a 70 but the average score (whether the mean, median, or mode) was 85, you did relatively poorly. If you made a 70 but the average score was only 55 then you did relatively well. In general, the significance of one observed value in a data set strongly depends on how that value compares to the other observed values in a data set. Therefore, we wish to attach to each observed value a number that measures its relative position.

Percentiles and Quartiles

Anyone who has taken a national standardized test is familiar with the idea of being given both a score on the exam and a “percentile ranking” of that score. You may be told that your score was 625 and that it is the 85th percentile. The first number tells how you actually did on the exam; the second says that 85% of the scores on the exam were less than or equal to your score, 625.

Definition

*Given an observed value x in a data set, x is the **P th percentile** of the data if the percentage of the data that are less than or equal to x is P . The number P is the **percentile rank** of x .*

Example:

A random sample of ten students is taken from the student body of a college and their GPAs are recorded as follows.

1.90 3.00 2.53 3.71 2.12 1.76 2.71 1.39 4.00 3.33

What percentile is the value 1.39 in the data set of ten GPAs? What percentile is the value 3.33?

Solution:

The data written in increasing order are

1.39 1.76 1.90 2.12 2.53 2.71 3.00 3.33 3.71 4.00

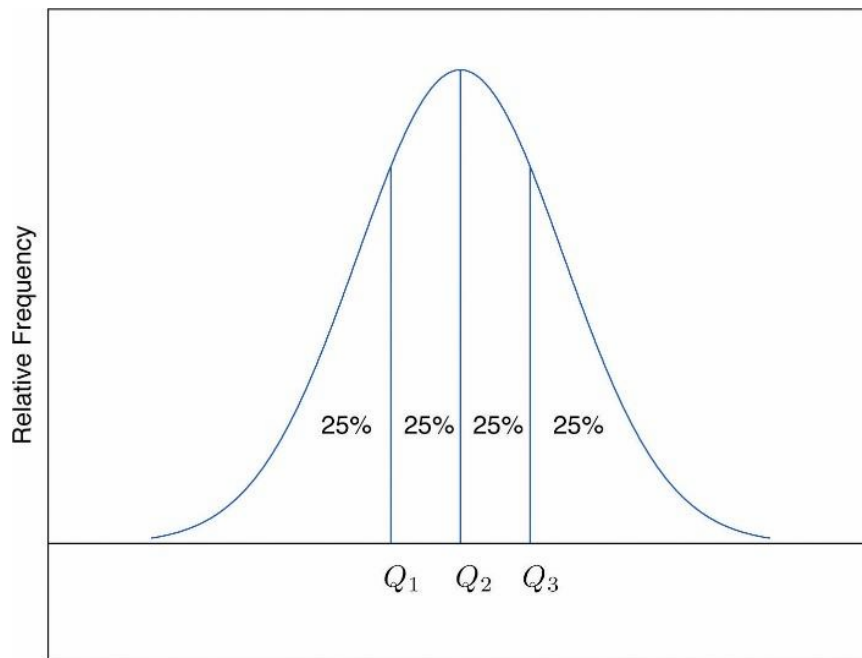


Figure 2.2 Data Division by Quartiles

Definition

For any data set:

1. The **second quartile** Q_2 of the data set is its median.
2. Define two subsets:
 1. the lower set: all observations that are strictly less than Q_2 ;
 2. the upper set: all observations that are strictly greater than Q_2 .
3. The **first quartile** Q_1 of the data set is the median of the lower set.
4. The **third quartile** Q_3 of the data set is the median of the upper set.

Example: Find the quartiles of the data set of GPAs

1.90 3.00 2.53 3.71 2.12 1.76 2.71 1.39 4.00 3.33

Solution:

First list the data in numerical order:

1.39 1.76 1.90 2.12 2.53 2.71 3.00 3.33 3.71 4.00

This data set has $n = 10$ observations. Since 10 is an even number, the median is the mean of the two middle observations: . Thus, the second quartile is $Q_2 = 2.62$. The lower and upper subsets are

Lower: $L = \{1.39, 1.76, 1.90, 2.12, 2.53\}$

Upper: $U = \{2.71, 3.00, 3.33, 3.71, 4.00\}$

Each has an odd number of elements, so the median of each is its middle observation. Thus, the first quartile is $Q_1 = 1.90$, the median of L , and the third quartile is $Q_3 = 3.33$, the median of U .

In addition to the three quartiles, the two extreme values, the minimum $x_{\min}$ and the maximum $x_{\max}$ are also useful in describing the entire data set. Together these five numbers are called the five-number summary of the data set:

$$\{x_{\min}, Q_1, Q_2, Q_3, x_{\max}\}$$

The five-number summary is used to construct a box plot as in Figure 2.3 "The Box Plot". Each of the five numbers is represented by a vertical line segment, a box is formed using the line segments at Q_1 and Q_3 as its two vertical sides, and two horizontal line segments are extended from the vertical segments marking Q_1 and Q_3 to the adjacent extreme values. (The two horizontal line segments are referred to as "whiskers," and the diagram is sometimes called a "box and whisker plot.")

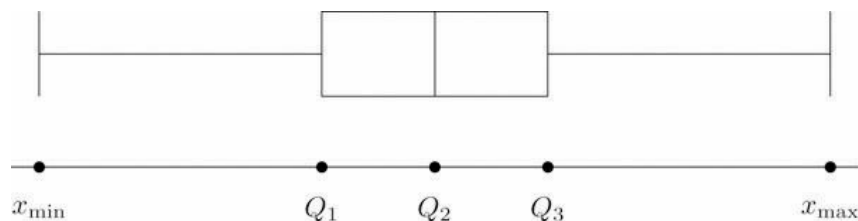


Figure 2.3 The Box Plot

Note that the distance from Q_1 to Q_3 is the length of the interval over which the middle half of the data range. Thus, it has the following special name.

Definition

The **interquartile range (IQR)** is the quantity

$$IQR = Q_3 - Q_1$$

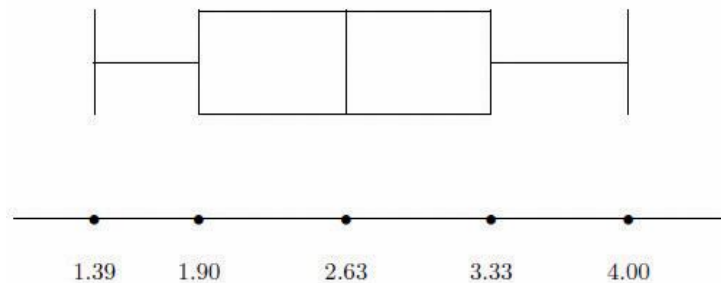
Example: Construct a box plot and find the IQR for the data below:

1.39 1.76 1.90 2.12 2.53 2.71 3.00 3.33 3.71 4.00

Solution:

$$x_{\min} = 1.39 \quad Q_1 = 1.90 \quad Q_2 = 2.62 \quad Q_3 = 3.33 \quad x_{\max} = 4.00$$

The box plot is



Summary

Measures of dispersion describe the spread of the data. They include the range, interquartile range, standard deviation, and variance. Measures of dispersion have two categories, the absolute and relative measures of dispersion. Absolute expresses scattering of observation in terms of distances, while relative compares distributions of two or more data sets.

The range is the difference between two extreme observations in the data set and the simplest measure of variability. Variance is the sum of the probabilities that various outcomes will occur multiplied by the squared deviations from the average of the random variable. Standard deviation is the measure of the average distance between the values of the data in the set and the mean. A percentile is a measure at which that percentage of the total values are the same as or below that measure. Quartiles are special percentiles. The first quartile, Q_1 , is the same as the 25th percentile. The second quartile, Q_2 , is the same as the 50th percentile / median. The third quartile, Q_3 , is the same as the 75th percentile. The interquartile range is the difference between the third quartile and the first quartile.